

Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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Chapter 1

Introduction

Chapter 2

Literature Review

2.1 Biomedical Information Extraction

2.1.1 Definition

Biomedical Information Extraction (IE) is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system: extracted entities and relations are later assembled into a knowledge graph and validated against biomedical ontologies **[NEED REFERENCE]** (section)

2.1.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with an analysis by Lu et al. (2011) demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data - a 2024 study mapping over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to a level where the volume of biomedical publications has made it difficult for researchers to keep track of new literature [2]. Literary analysis is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and a substantial majority of literature goes unseen that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this thesis, as a corpus that is both large and continuously updated is exactly the setting in which static benchmark-trained pipelines are most likely to be used.

2.1.3 Tasks

Biomedical IE is usually comprised of several sub-tasks: Named Entity Recognition (NER), which identifies and stores types of spans of text referring to biomedical concepts, and Relation Extraction (RE), which identifies semantic relationships between recognised entities, and Named Entity Normalisation (NEN), which maps extracted mentions to pre-defined identifiers from an existing ontology [4, 5].

Unlike NER, which only labels a span of text as belonging to a given entity type, NEN maps that span to an established identifier from a pre-defined ontology or controlled vocabulary, storing which specific concept the span is referring to, rather than what kind of concept it is. This resolves terminology ambiguity challenges discussed in the following section.

2.1.4 Challenges

Long or complex sentences, which are common in biomedical abstracts, summarising multiple findings into a single sentence, can cause a significant challenge for these tasks [6]. This is particularly prominent for relation extraction, which typically requires attention over the text’s distance between the two entities within a sentence: if two entities are far apart in this text, multiple clauses and mentions of competing entities can make it more difficult to correctly associate the entities with their corresponding relation [7], increasing the risk of both False Negatives (FNs) and False Positives (FPs). Sequential relation extraction models have been shown to have difficulty capturing these long-range dependencies, resulting in poorer performance on longer sentences [8]. The information density of biomedical literature may partly reflect conventions of scientific communication, which emphasise conciseness, clarity and precision of information [9].

Beyond task complexity, biomedical text includes domain-specific challenges that separate it from general IE: dense and evolving terminology [10, 11], and high rates of entity name ambiguity [12], where the same underlying concept is frequently expressed through several distinct strings, such as a drug’s brand name versus its generic name, or a gene’s full name versus a locally-defined abbreviation within a specific paper, while on the other hand, a single string can refer to different concepts depending on context. As a result, extracted entities referring to the same concept would be treated as distinct, fragmenting information than consolidating it. NEN aims to address this, mapping ambiguous or synonymous mentions onto a shared established identifier. This task is largely important for subsequent knowledge graph construction, as graph nodes rely on identifiers to be queryable and to allow separate mentions of the same concept to be merged into a single node [NEED REFERENCE].

In addition to these domain-specific challenges, significant methodological limitations compound the difficulty of RE evaluation: meaningful comparison

between biomedical IE systems requires consistent evaluation frameworks, while strong generalisation across literature distributions requires models to perform reliably across heterogeneous data [NEED REFERENCE][MORE DEPTH].

2.1.5 Traditional Methods and Evaluation

Biomedical IE has progressed from rule-based systems towards feature-based machine learning and deep learning approaches built on pretrained transformer models. Throughout this progression, development and evaluation have been mainly driven by benchmark datasets, with models commonly trained and evaluated on fixed, established corpora. However, relying on individual benchmarks introduces the comparability and generalisability limitations outlined previously. Recent work has therefore aimed to address these limitations through more consistent evaluation frameworks and broader, heterogeneous training data: Sanger et al. (2025), for example, compared multiple pretrained language models before evaluating knowledge extraction of the best-performing model across five datasets within a consistent evaluation framework [13]. Similarly, Lai et al. (2023) introduce BioREx, which harmonises heterogeneous biomedical RE datasets to support broader training and improve generalisation [14].

Both of these approaches, however, rely on Pretrained Language Models (PLMs) tuned on labelled benchmark data, with performance assessed against in-distribution, fixed, annotated datasets. This highlights an important methodological limitation for practical application, where extraction systems must perform robustly on dynamic, contemporary literature, rather than a static benchmark corpus. Therefore, the extent to which performance established on benchmark data transfers to unseen contemporary literature remains an important question and a motivator for Research Question 1 (RQ1).

2.2 Large Language Models for Biomedical Information Extraction

2.2.1 Transition from Specialised Models to LLMs

More recently, Large Language Models (LLMs) have driven a broader shift in knowledge extraction and knowledge graph construction from rule-based and module pipelines towards generative and adaptive frameworks [15]. Early evidence supports competitive extraction performance with Zhang et al. (2024) reporting an F1 score of 0.84 for GPT-4 on biomedical RE for the Genetic Association Database (GAD) benchmark dataset, with GPT, BioBERT, and PubMedBERT performing comparably on the GAD dataset under certain extraction system frameworks [16]. However, it is important to note that GPT-3.5 and BioBERT v1.1 performed considerably worse than PubMedBERT on the

masked ChemProt dataset, suggesting that relative model performance may vary substantially across biomedical RE tasks and datasets.

Additionally, performance gains from LLMs are not limited to biomedicine. Evaluations of LLMs for knowledge graph construction have observed an F1 score of 99.35% for Meta’s LLaMA 3 model on OpenStack system log data [17], indicating extraction capability is not domain-specific.

This body of evidence motivates the pipeline design adopted in this project: an LLM-based extraction approach using structured JSON outputs and few-shot prompting, developed on an existing benchmark, BioRED [18], before evaluation on a newly-assembled contemporary corpus.

2.2.2 Prompt-Based and Few-Shot Extraction

Unlike specialised pretrained language models such as BioBERT and PubMedBERT, which require task-specific fine-tuning on labelled data to perform NER or RE, generative LLMs can perform biomedical IE directly through prompting, without any parameter tuning [19]. This approach can allow flexibility through natural-language instructions, rather than requiring additional model training. As a result, a single general-purpose model can potentially be adapted to different entity types, relation schemas, and extraction tasks through only prompt refinement [NEED REFERENCE].

Several approaches can be implemented: zero-shot prompting - where the model is given only a natural-language description of the task, few-shot prompting - where the prompt includes structured examples of exemplary outputs corresponding to given inputs, and schema-guided instruction, where the prompt specifies the exact entity and relation types that the model should extract [20, 21, 22].

The extent to which an LLM is able to compete with fine-tuned extraction systems depends heavily on how the prompt itself is constructed. Recent comparisons between zero-shot and few-shot prompting on biomedical IE tasks suggest that including examples in the prompt can improve extraction quality, as can annotation guidelines or provided schema within the prompt itself [21]. However, these gains are not consistent across biomedical IE tasks: Zhang et al. (2024) observed only limited improvements from one-shot prompting on some datasets, with performance decreasing on the ChemProt dataset. Given ChemProt’s more complex relation schema [23], these findings suggest that the effectiveness of few-shot prompting may depend on task complexity and how well the selected examples represent the target annotations [NEED REFERENCE]. This prompt-dependency is important for this project specifically as it provides a basis for developing and iterating on multiple Extraction Pipeline Iterations (EPI) during BioRED-based development, rather than treating prompt design as a single, fixed choice that remains unexamined.

2.2.3 Performance and Limitations

Despite this flexibility, LLM-based extraction introduces failures that are usually absent from, or less significant in fine-tuned supervised models. The most widely discussed is drawing from domain knowledge: LLMs may produce extracted entities or relations that are plausible but not present in the supplied text [22]. This is particularly important for biomedical IE, where an extracted relation is only useful if it is grounded in the given literature, rather than drawn from existing background knowledge.

Problems also exist with LLM consistency and control: LLM outputs can vary across repeated runs on the same input [24], and extraction quality has been shown to be sensitive to changes in prompt wording or the specific examples chosen for few-shot prompting [16, 21]. Models may also fail to strictly follow a predefined relation schema, extracting relation types outside the intended ontology or omitting required output fields, which can complicate downstream pipelines that assume a fixed schema [24]. Together, this behaviour may create a precision-recall trade-off where more exhaustive extraction increases recall at the cost of precision, while more conservative prompting tends to do the opposite.

2.2.4 Implications for Biomedical Information Extraction

2.3 Knowledge Graph Construction Using Large Language Models

2.4 Benchmark Datasets and Evaluation

2.5 Generalisation to Contemporary Biomedical Literature

Chapter 3

Methodology

3.0.1 Research Design

3.0.2 Datasets

3.0.3 System Architecture

3.0.4 Information Extraction Pipeline

3.0.5 Pipeline Development

3.0.6 Contemporary Knowledge Graph Construction

3.0.7 Evaluation

3.0.8 Cost Analysis

3.0.9 Software Implementation

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

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